

DermaScan AI Architecture Profile

DermaScan AI is a high-fidelity, hybrid clinical screening platform built to help patients detect skin conditions instantly, track healing progress over time, consult an AI triage chatbot, and manage sun safety. The system is split into a Flutter frontend and an ensembled deep learning Flask backend.

1. Core Architecture & Tech Stack

Frontend (Flutter & Dart):

- Design: Dark theme (Obsidian background #0A0E1A, Surface card #111827) with Mint Teal and Electric Violet accents.
- State & UI: Features custom sliders, circular countdown timers, and API connectors.

Backend (Flask & Python):

- REST API: Handles scan uploads, database entries, chatbot sessions, and UV computations.
- AI Engine: Uses TensorFlow to serve deep learning models and output Grad-CAM attention heatmaps.

Database (SQLite - diary.db):

- users: Stores user login credentials, profiles, and phone numbers.
- diary_entries: Stores scan diagnostics, confidence values, base64 images, and progress logs.

2. The Seven Pillars of User Features

* I. Ensemble AI Classifier

Combines MobileNetV3Large (for fast 230ms inference) and EfficientNetB3 (for Grad-CAM++ pixel-attention mapping) to diagnose 10 classes of skin conditions (Melanoma, Eczema, Moles, BCC, etc.).

* II. Real-Time UV Index Advisor

Calculates solar decay curves peaking at noon based on coordinates. Provides a dashboard meter, safe exposure timer, clothing guides, and 5-hour forecasts.

* III. AI Triage Chatbot (DermaChat)

Performs preliminary conversational triaging (High/Moderate/Low risk) using regex keyword boundaries and provides suggest chips.

* IV. Treatment Tracker & Swiper

Allows comparison of lesion progress side-by-side using an interactive slider-clipper and charts severity curves over time.

* V. Clinical PDF Report Export

Generates styled PDF briefs with medical details, captured photos, and official disclaimer notices for sharing with dermatologists.

* VI. Obsidian Map Locator

A vector-painted locator widget mapping hospitals and clinics with distance, ratings, and phone calls without API bills.

* VII. Dual Auth System

Supports credential-based logins and passwordless phone number OTP auth.

3. Project Structure & Key Files

- backend/app.py: Flask endpoints, AI model preloading, chatbot, and UV calculations.
- lib/services/api_service.dart: Network API connector connecting screens to Python.
- lib/screens/home_screen.dart: Shell home layout, UV gauge painter, and Sun Safety sheet.
- lib/screens/chatbot_screen.dart: Conversational triage chat interface.
- lib/screens/treatment_progress_screen.dart: Visual comparison slider view.
- lib/services/pdf_report_helper.dart: A4 PDF exporter for patient clinical briefs.